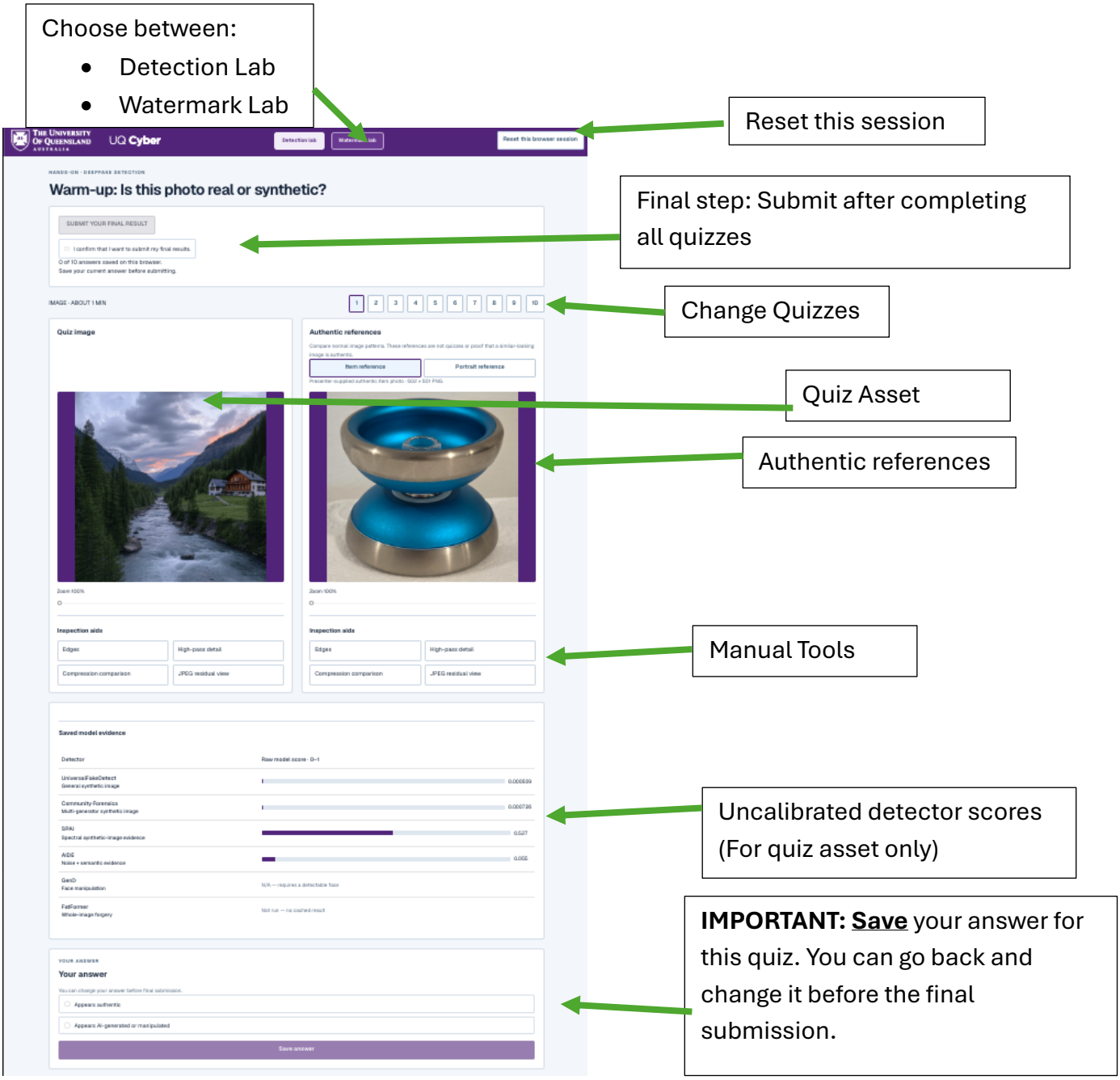
UQ Cyber Research Centre

Deepfake Identification – Hands-on Session

Before the session: Please enter your 'nickname' to track your score at the end of the session. Please try to use some unique name or add some random numbers at the end.

During the session: In this hands-on session, you will experience some short challenges in identifying whether each media asset presented is authentic, edited using AI or manual techniques, fully synthetic, or a mixture of the latter two. We have provided various tools to help you assess its authenticity.

- **Authentic References:** Authentic images that provide same set of tools.
- **Manual Tools:** Simple image inspection tools that help you examine whether an asset may have been modified.
- **Automated Detectors:** We provide results from a variety of publicly available AI-based detectors. The scores are precomputed, so you won't need to run the detectors in real time.



Using the Inspection Tools

We have provided several manual tools as inspection aids, which you can quickly apply to the media and compare with the original. For videos, these tools inspect a pre-selected still frame. These views alone cannot establish authenticity. Please note that these possible clues can also exist in authentic video/photos.

1. **Edges:** Select Edges and adjust Amplification to make outlines easier to see. Compare hair, object borders and fine lines with the original. Bright edges show strong changes in brightness. Poor blending, imperfect alignment and sharpening may make these edges brighter. Possible clues to examine include:
    - An unexpected line around a region such as the cheek or jaw.
    - Doubled or broken outlines near the eyes, lips or other facial features.
    - A very sharp face outline surrounded by softer details, or the reverse.
    - Bright halos or exaggerated outlines around facial features.
2. **High-pass detail:** Select High-pass detail and adjust Detail scale and Amplification. This highlights differences between the image preview and a blurred copy, making edges, fine texture and noise easier to examine. Differences worth examining include:
    - Nearby skin areas with unexpectedly different amounts of fine detail.
    - An overly smooth face surrounded by more detailed skin, with an abrupt change in fine texture or noise.
    - Unusually grainy or strongly sharpened texture in one region.
3. **Compression comparison:** Choose Low-quality JPEG or Reduce dimensions to ¼, then compare with the Reference preview. This shows how processing can create or hide visible clues, rather than directly detecting editing. Examples include:
    - Blocks or distorted edges appearing after JPEG compression.
    - Fine skin texture, hair or small text becoming less clear than other parts of the image.
    - A suspicious boundary becoming harder to see after resizing.
4. **JPEG residual view:** Adjust JPEG quality, Amplification or Colour channel, then select Recalculate. This highlights differences between the preview and a newly JPEG-compressed copy. Compare similar areas and look for:
    - A region that reacts differently from nearby areas with similar texture.
    - An abrupt change in the residual pattern across a possible editing boundary, such as around the cheeks, jaw, eyes or mouth.
    - Bright or purple patterns can result from ordinary texture, edges and compression, these patterns alone do not establish that a region was edited or AI-generated.

Uncalibrated scores: The detector scores have not been validated as reliable probabilities for these workshop cases. They are raw model outputs for each file, not measurements against the training images. A score of 0.8 does not mean an 80% chance of being fake. Reliable calibration requires representative validation data that covers relevant, up-to-date examples.

However, these scores can still provide supporting clues, although they depend on what each model learned during training. For the six 0–1 image-classifier scores, higher values lean towards that model's synthetic or face-manipulated class. Video methods use their own scales and comparisons. Do not average different detectors or treat them as votes. A low score is not proof of authenticity.